POCKETSMART AI

Project Documentation

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Bachelor of Computer Applications (BCA)

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Academic Year

2026-2027

Introduction

About the Project

PocketSmart AI is a smart budget planning application developed to help users make cost-effective decisions for different needs. The system provides recommendations based on the user's budget and selected preferences.

The application includes three planning modules:

Home Interior Planner

Party Planner

Jewelry Planner

Users can enter their budget and preferences to receive suitable recommendations.

Why This Project Was Developed

Many people find it difficult to plan events, interior decoration, or jewelry purchases within a limited budget. PocketSmart AI was developed to simplify this process by providing quick and budget-friendly suggestions.

Objectives

The main objectives of PocketSmart AI are:

To help users plan within their budget.

To provide quick recommendations.

To reduce planning time.

To improve decision-making.

To create a simple and user-friendly platform.

How It Helps Users

Saves time.

Reduces unnecessary expenses.

Provides organized planning.

Easy to use for everyone.

Problem Statement

People often struggle to:

Manage expenses properly.

Select suitable interior items.

Plan parties within budget.

Choose jewelry according to their financial limits.

Without proper planning, users may spend more than expected.

Therefore, a budget-based planning solution is required.

Proposed Solution

PocketSmart AI provides a centralized platform where users can enter their budget and requirements.

The system analyzes the user input and generates recommendations for:

Home Interior Planning

Party Planning

Jewelry Selection

This helps users make informed and budget-friendly decisions.

Features / Modules

Home Interior Planner

User enters budget.

Selects room type.

Receives interior recommendations.

Party Planner

User enters budget.

Enters number of guests.

Selects event type.

Receives party planning suggestions.

Jewelry Planner

User enters budget.

Selects occasion.

Selects style.

Uploads jewelry image (optional).

Receives jewelry recommendations.

User Authentication

Registration

Login

Dashboard Access

Recommendation History

View previous recommendations.

Technologies Used

Technology Purpose

HTML Structure of web pages

CSS Styling and design

JavaScript Client-side interactions

Python Backend programming

FastAPI Web framework

Jinja2 HTML templating

Uvicorn ASGI server

System Requirements

Hardware Requirements

Processor: Intel Core i3 or above RAM: 4 GB minimum

Storage: 500 MB free space

Software Requirements

Windows 10/11

Python 3.10+

Visual Studio Code

Google Chrome

Python Requirements

FastAPI

Uvicorn

Jinja2

Python Standard Libraries

System Design / Working

Working Process

1. User registers or logs in.

2. User accesses dashboard.

3. User selects planner.

4. User enters budget and preferences.
5. FastAPI processes the request.

6. Recommendation is generated.

7. Result is displayed to the user.

Flowchart

Start

|

User Login/Register

|

Dashboard

|

Select Planner

|

Enter Budget & Preferences

|

Process Request

|

Generate Recommendation

|

Display Result

|

End

Screenshots

Figure 1: Registration Page



Figure 2: Login Page



Figure 3: Dashboard Page

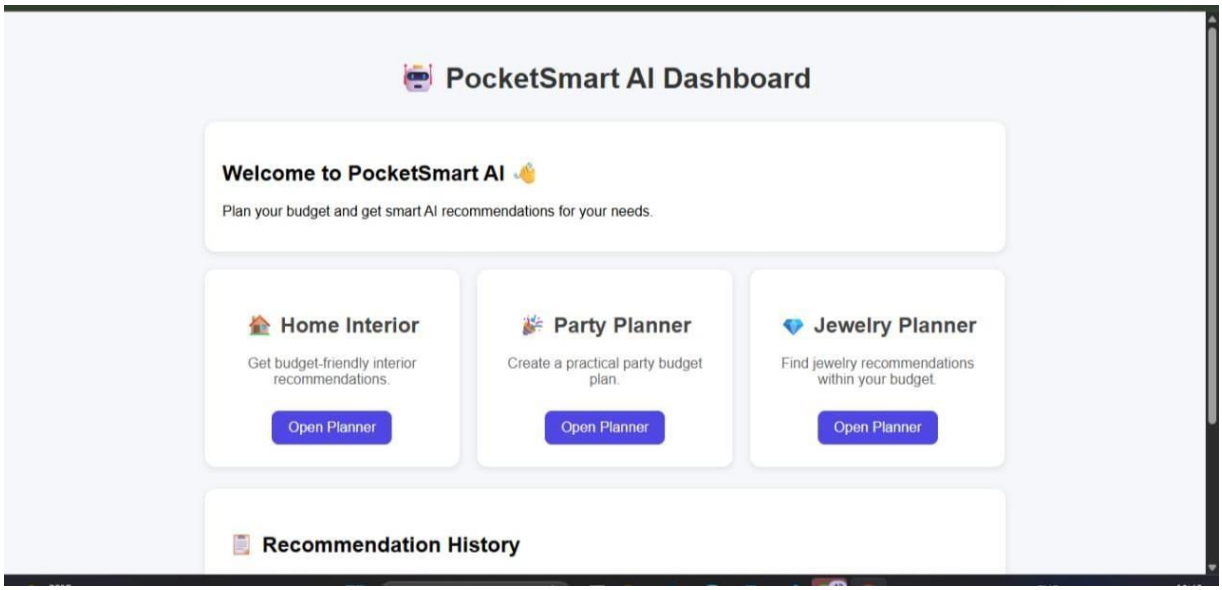


Figure 4: Home Interior Planner

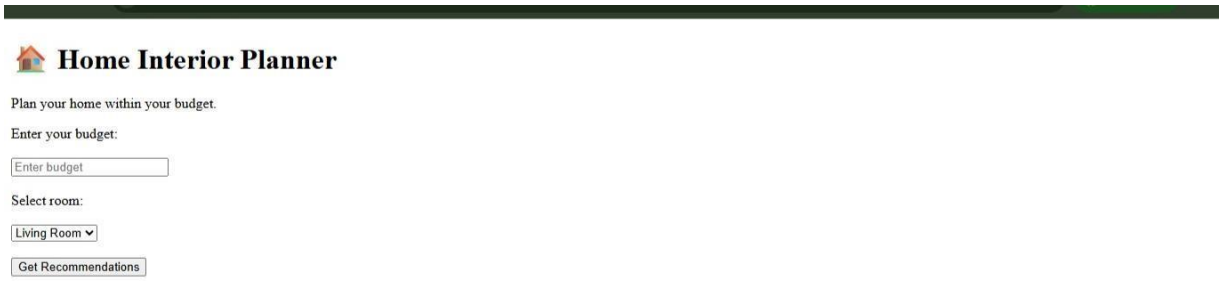


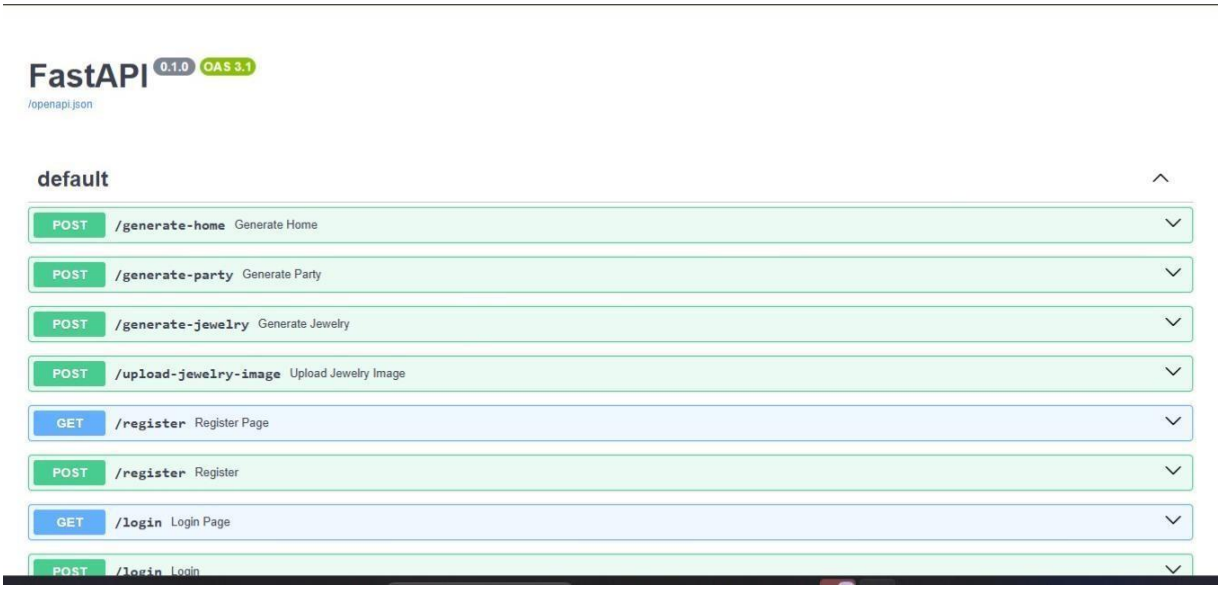
Figure 5: Party Planner



Figure 6: Jewelry Planner



Figure 7: FastAPI Swagger Documentation



Project Structure

PocketSmart-AI/

```
|
├── main.py
├── requirements.txt
├── history.json
|
├── templates/
|       ├── register.html
|       ├── login.html
|       └── dashboard.html
```

```
│       ├── home.html
│       ├── party.html
│       ├── jewelry.html
```

Important Files

Main.py

Contains FastAPI backend logic.

Templates/

Stores HTML pages.

Static/

Stores CSS, JavaScript and images.

Testing

Test Case Input Expected Output Actual Output

User Registration Username & Password Account Created Passed

User Login Valid Credentials Login Success Passed

Home Planner Budget & Room Recommendations Passed

Party Planner Budget & Guests Recommendations Passed

Jewelry Planner Budget & Style Recommendations Passed

Advantages

Easy to use.

Budget-based recommendations.

Simple interface.

Multiple planning modules.

Fast response.

User-friendly design.

Limitations

Uses predefined recommendation logic.

Recommendations depend on the availability of the Gemini API.

Limited recommendation categories.

No database integration.

Limited personalization.

Future scope

Future improvements include:

Database Integration

Database connectivity.

Personalized recommendations.

More planning categories.

Mobile application development.

User profile management.

Advanced analytics.

Conclusion

PocketSmart AI is a simple and effective budget planning application that helps users make informed decisions regarding home interiors, parties, and jewelry purchases. The system provides a user-friendly interface and generates useful recommendations based on the user's budget. This

project demonstrates the practical implementation of FastAPI and web technologies in solving real-world planning problems.

References

1. Python Documentation

2. FastAPI Documentation

3. HTML Documentation (W3Schools)

4. CSS Documentation (W3Schools)

5. JavaScript Documentation (MDN)

6. Visual Studio Code Documentation

7. Uvicorn Documentation